

BASELINE TRIAGE MODEL - RESULTS REPORT

To: Dr De Freitas; Dr Marcus Reyes, Consultant Emergency Physician

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Re: Two baseline classifiers for ED triage (ESI), and why recall for Level 1 is the metric that matters

Setup: 55,121 patients, 80/20 stratified split, random seed 42, outcome column (disposition) excluded to prevent leakage

1. Results

Two simple models were trained and compared against a stratified random guess. Logistic regression is the stronger baseline. The decision tree, even bounded at depth 6 for interpretability, collapses to predicting the common ESI 3 band and never flags a Level 1 patient.

Model	Accuracy	Macro F1	Weighted F1	ESI-1 recall
Random guess (stratified)	0.375	0.204	0.375	0.00
Logistic regression	0.685	0.518	0.679	0.25
Decision tree (depth 6)	0.560	0.226	0.479	0.00
Logistic reg. (class-weighted)	0.585	0.428	0.611	0.69

Both real models beat the coin flip on accuracy, so the data carries genuine signal. But accuracy is the wrong headline, as the next section explains.

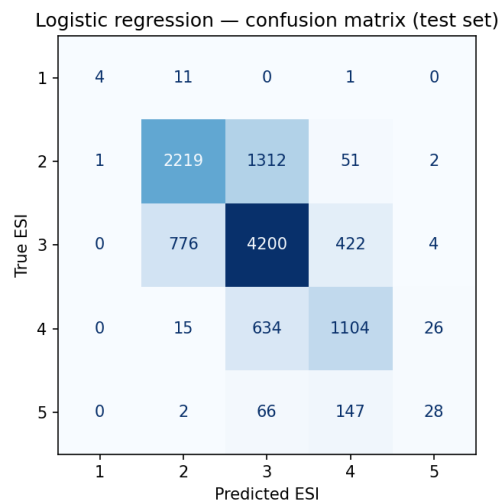


Figure 1. Logistic regression confusion matrix. The top row (true ESI 1) is the concern: most Level 1 patients are predicted as Level 2 or lower.

2. Primary metric and justification

The primary metric is recall on the urgent classes, above all ESI 1, which is equivalent to keeping the under-triage rate of critical patients low. In triage the two error types are not equal. Sending a Level 1 patient to the waiting room can be fatal; sending a Level 4 patient to a resus bay wastes a bed. Overall accuracy treats these the same and is dominated by the large ESI 3 group, so a model can score 0.68 while missing three quarters of the sickest patients. Recall on ESI 1 measures exactly the failure we cannot tolerate. Macro F1 is the secondary metric because it weights every acuity level equally, unlike weighted F1 (0.68) which is inflated by the common classes; the gap between the two, 0.52 against 0.68, is itself the signal that the model is weak where it matters.

3. Beyond the baseline

Four additional analyses were run to test the model rather than just score it.

Safety decomposition. Every prediction was labelled as over-triage, correct, or under-triage. Logistic regression under-triages 38% of the sickest (ESI 1-2) patients into a non-urgent band; the decision tree under-triages 66%. This is the number a clinician actually reads.

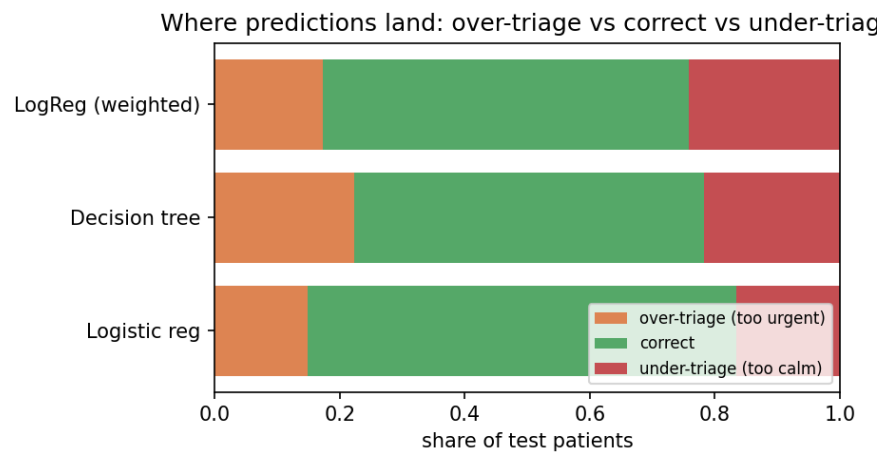


Figure 2. Where predictions land. Red (under-triage) is the clinically dangerous portion.

The recall lever. Setting class weights to balanced roughly triples ESI-1 recall, from 0.25 to 0.69, and cuts dangerous under-triage of the sickest patients from 38% to 28%, in exchange for lower overall accuracy (0.685 to 0.585). This trade, catching more critical patients at the cost of more false alarms, is a decision for the ED Board, not one the model should make silently.

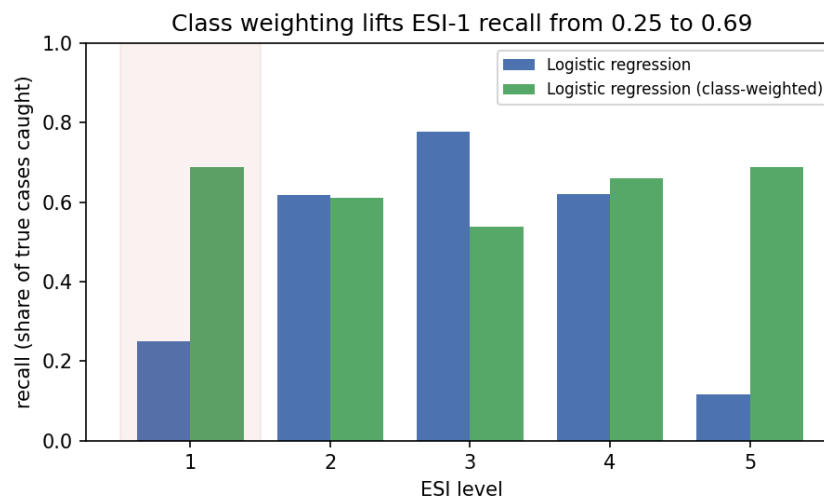


Figure 3. Class weighting lifts recall on the rare, urgent classes (ESI 1 and 5).

How reliable is that recall? With only 16 ESI-1 patients in the test set, the recall estimate is shaky. A 2,000-sample bootstrap puts the 95% confidence interval at roughly 0.06 to 0.50. We genuinely cannot yet measure Level 1 recall precisely, which argues for more data and caution before trusting any single figure on the rarest class.

Does the model agree with physiology? Checking the feature directions in an urgent-versus-not model, every key feature points the clinically correct way: altered mental status, chest pain, shortness of breath, higher heart and respiratory rate, older age and lower oxygen all raise urgency, while sore throat and dental pain lower it. The model learned real clinical patterns, not artefacts.

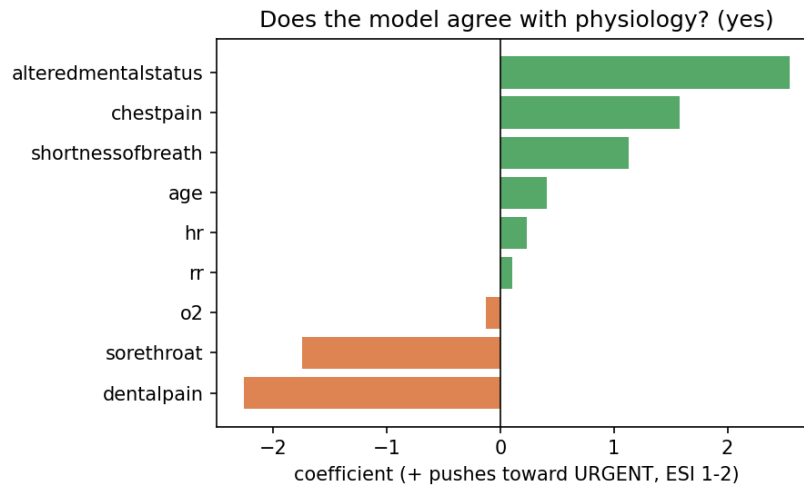


Figure 4. Feature directions in an urgent (ESI 1-2) model. Green raises urgency, orange lowers it, all as a clinician would expect.

4. Failure-mode reflection

The failure mode I am most worried about is under-triage of ESI 1. The plain baseline misses three out of four Level 1 patients. In a real ED that is a critically ill person, an early myocardial infarction or sepsis, routed to wait instead of to resuscitation, where minutes change outcomes. Class weighting reduces the miss rate to roughly one in three, but does not remove it, and the wide confidence interval means we cannot yet trust even that. The honest conclusion for the Board is that this baseline clears the agreed bar, it beats a coin flip and behaves in clinically sensible ways, but it is nowhere near safe to deploy. It is a starting point that tells us where the effort must go: the rare, urgent patients.